



Syllabus

Applied Machine Learning

Course Introduction

Assistant Professor | **Chun-Hsiang Chan**

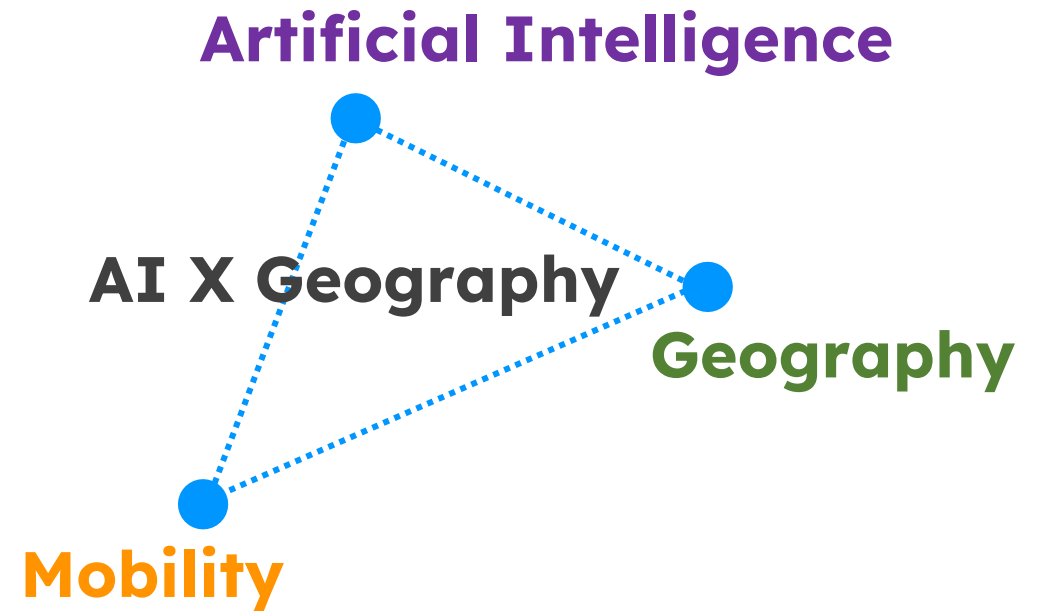
Department of Geography
National Taiwan Normal University



Outline

#GeoAI #Mobility #Geography #STDS

- 01 About CCH
- 02 Course Intro
- 03 Why do you need to take this course?
- 04 What will you learn from this course?
- 05 Grading Policy
- 06 Question Time



詹竣翔博士

#Dr. Chun-Hsiang Chan
#GeoAI #Medicine #Chemistry

01 現職 (2023/8~)

專任助理教授 | 國立臺灣師範大學地理學系

02 主要學歷

博士 | 國立臺灣大學地理環境資源學系 (15-21)

碩士 | 實踐大學食品營養與保健生技學系 (18-20)

碩士 | 國立臺灣大學地理環境資源學系 (13-15)

學士 | 國立臺北教育大學社會與區域發展學系 (09-12)

專題生 | 國立臺灣大學化學系[有機金屬] (10-12)

03 主要工作經歷

專任助理教授 | 中原大學智慧運算與大數據學系 (22-23)

AI長 | 台灣資安鑄造股份有限公司 (22-23)

兼任助理教授 | 淡江大學人工智慧學系 (21-22)

博士後研究員 | 臺北醫學大學醫學系放射線學科 (21-22)

博士後研究員 | 臺北市立萬芳醫院影像醫學部 (21-22)

資料科學家 | 財團法人資訊工業策進會 (17-19)

04 主要研究方向

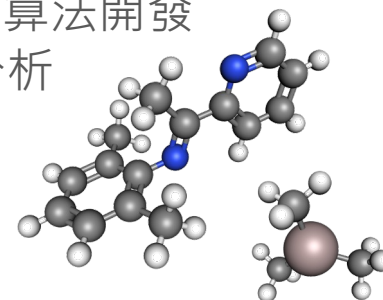
#1 地理人工智慧與演算法開發

#2 時空大數據資料分析

#3 複雜網路分析

#4 臨床醫學分析

#5 基礎與應用化學



05 [招生中] 大學專題生/碩士生



More info.



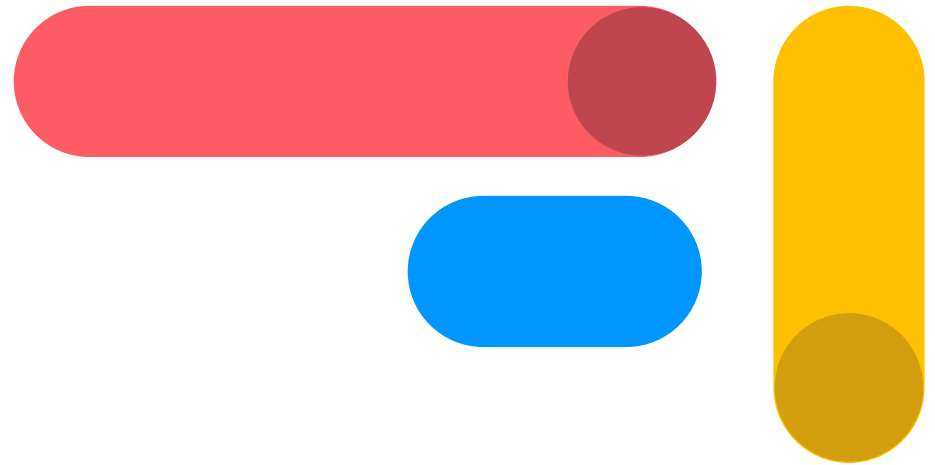
Course Intro.

- In the era of data-driven research and intelligent systems, one is increasingly confronted with the challenge of extracting meaningful patterns and actionable insights from complex datasets.
- This course introduces the fundamental theories and practical tools of Machine Learning and Deep Learning, covering topics such as classification, clustering, convolutional neural networks, sequential models, natural language processing, large language models, graph neural networks, and reinforcement learning.

Course Intro.

- The course will be conducted through a combination of theoretical instruction and hands-on practice, enabling students to understand core learning algorithms while gaining practical experience in model implementation, evaluation, and application.

Course Intro.



Week	Date	Content
1	Sep. 7	Foundation of Machine Learning
2	Sep. 14	Classification (I)
3	Sep. 21	[Online] Classification (II)
4	Sep. 28	National Holiday
5	Oct. 5	Clustering
6	Oct. 12	Foundation of Deep Learning
7	Oct. 19	Convolutional Neural Network (I)
8	Oct. 26	National Holiday
9	Nov. 2	[Online] Convolutional Neural Network (II)
10	Nov. 9	Sequential Models

Week	Date	Content
11	Nov. 16	Natural Language Processing (I)
12	Nov. 23	Natural Language Processing (II)
13	Nov. 30	Large Language Models
14	Dec. 7	Graph Neural Networks
15	Dec. 14	Reinforcement Learning
16	Dec. 21	Final Report

Why do you need to take this course?

In the era of artificial intelligence and data-driven decision-making, machine learning has become an essential capability for analyzing complex data, building predictive models, and solving real-world problems.

This course provides the theoretical foundation and practical experience needed to understand how intelligent systems learn from data and how such techniques can be applied across different domains.

What will you learn from this course?

Through this course, students will learn the core concepts and methods of Machine Learning (ML) and Deep Learning (DL), including classification, clustering, convolutional neural networks, sequential models, natural language processing, large language models, graph neural networks, and reinforcement learning.

By combining theoretical lectures with hands-on practice, students will gain experience in model implementation, evaluation, and application, and will integrate their learning through a final project report.

Grading Policy



All you have to do is study hard and feel free to ask question when you do not understand.



I believe that if you fulfill all required items, and then you will pass this course / get a high GPA.

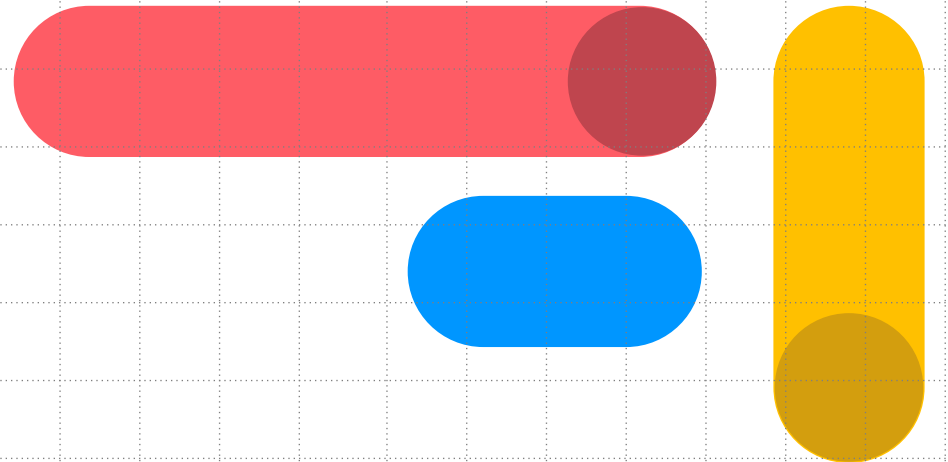


Do not worry about the grade! The most important things is what you learn from this course.

Attendance 10%

Assignment 60%

Final Report 40%



The End

Thank you for your attention!

Email: chchan@ntnu.edu.tw

Website: <https://toodou.github.io/>